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Received: 4 May 2024 | Revised: 6 December 2024 | Accepted: 6 December 2024

DOI: 10.1002/btm2.10742

REVIEW ARTICLE

BIOENGINEERING & TRANSLATIONAL MEDICINE

Recent developments in microwire-structured intracortical electrode arrays for brain–machine interfaces

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Funding information
Australian Research Council, Grant/Award Numbers: DE220100302, FT190100215

Abstract

Brain–machine interfaces (BMIs) have experienced remarkable advancements in recent years, marked by multiple companies initiating human trials. Consequently, the interface between the brain and electrodes has become more critical than ever, requiring implanted electrodes to be not only biocompatible and minimally invasive but also capable of remaining functioning in the brain for a lifetime. While significant progress has been made in the manufacturing of intracortical electrodes, challenges persist in ensuring longevity and minimizing tissue damage. Additionally, the reliance on manual labor in fabrication techniques poses obstacles to large-scale production for commercialization. In this review, we explore recent breakthroughs and obstacles in the fabrication of microwire-structured electrode arrays, wherein single wires are arranged in an *xy* matrix for cortical penetration. We discuss the impact of various fabrication strategies and materials on implant longevity, as well as the remaining challenges in this field.

KEYWORDS

brain implants, brain–machine interfaces, cortical implants, flexible electronics, microelectrodes, neurons, recording, sensing

Translational Impact Statement

Recent advancements in brain–machine interfaces (BMIs) have accelerated human trials, emphasizing the critical need for biocompatible, minimally invasive, and long-lasting implanted electrodes. This review delves into innovative microwire-structured electrode arrays, offering insights into fabrication techniques and materials to enhance implant longevity and minimize tissue damage. Overcoming these challenges could revolutionize BMIs, enabling widespread commercialization and improving the quality of life for individuals with neurological disorders.

1 | INTRODUCTION

With an increasingly aging population, the prevalence of degenerative brain diseases has increased,[1] and consequently, there is a pressing need for improved detection and treatment of neurological disorders. Recent advances in brain–machine interfaces (BMIs) have opened a new avenue for the diagnosis and treatment of these brain diseases, including Parkinson's disease,[2] epilepsy,[3,4] depression,[5] and obsessive-compulsive disorder.[6] These interfaces can establish direct communication from the brain to external electronics. Most of the current BMIs in

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clinic use external neural electrodes that do not penetrate into brain tissues, such as electroencephalography (EEG) and electrocorticography (ECoG). EEG utilizes large electrodes placed on the surface of the scalp and provides broad information about brain activity across the whole brain but minimal at the cellular level.[7] ECoG electrodes are placed in direct contact with the cortex of the brain; the closer proximity to the brain offers a better signal-to-noise ratio with higher spatial resolution at the cellular level compared to EEG. In contrast to the non-invasive EEG electrodes, ECoG is invasive, therefore, implantation of ECoG electrodes is riskier and more traumatic.[7] Traditionally, it requires the removal of large sections of the patients skull to accommodate the matrix of large implantable electrodes (in the order of cms).[8] Recent strides in implantable electrode design have concentrated on increasing spatial resolution throughout the brain while maintaining a small size and adequate neuron coverage for more robust signal interpretation, as highlighted by recent studies from Hong et al. and Vatsyayan et al.[9,10] Smaller implantable electrodes have the potential to reduce surgical risk and tissue trauma. A smaller section of the skull can be delicately excised to facilitate implantation, further enhancing spatial resolution at the cellular level, which enables the monitoring of single-cell activity.[11] Applications such as prosthetic control and speech-to-text functionality demand heightened resolution and electrode count for practical efficacy.[12–14] These enhanced resolution signals need to harness the power of contemporary computing, opening doors to the development of more efficacious interventions. By interfacing with neural signals at the single neuron resolution, we pave the way for an unparalleled understanding of human brain function.

To address these resolution needs, the deployment of an intricate matrix of minuscule electrodes emerges as a promising solution. Such a matrix can be strategically distributed across the brain's cortex, providing exceptional spatial resolution both at the macro and cellular levels. This advancement holds the key to delving deeper into the intricacies of cellular and neuronal dynamics within the brain, a pivotal foundation for future endeavors in comprehending and predicting brain activity and consciousness.[15] Currently, there are many implantable arrays of electrodes that are commercially available for this purpose, but only a few of them have been implanted in humans. Most prominent is the Utah array from Blackrock Neurotech, widely used in animal research and clinic.[14,16] This array with electrodes of diameters in the range of tens of μm level is able to record single units across hundreds of channels. However, Utah arrays are composed of silicon needles, whose large size and inherent stiffness can produce extensive scarring of brain tissue which often leads to failure of the implant long term. Thus, a significant portion of scientific research is directed toward the development of arrays featuring thinner and more flexible microwire electrodes that better conform to brain tissue, reducing inflammation, and ensuring sustained high performance over extended periods.[17] To make an implantable neural electrode suitable for human application, long-term durability is essential. The ultimate objective of neural interfacing is to have a minimally invasive implant capable of single-cell resolution communication while seamlessly interfacing with a patient's tissue throughout their lifetime.

Microwire-structured electrodes possess geometries and physical properties that are promising for achieving long-term neural interfacing. Notably, the *xy* structure allows the dura to be sutured over the electrode arrays, reducing the likelihood of implant rejection. However, fabricating such arrays comes with its own challenges, as microwires need to be assembled and functionalized individually. In this context, we will discuss recent advancements in materials and manufacturing technologies that are specific to this three-dimensional configuration and enable the fabrication of high-channel-count microwire electrode matrices. For clarity, we define a microwire as a single wire with a diameter of less than 50 μm. A microwire electrode array (MwEA) refers to an assembly of microwires arranged in a matrix configuration in the *xy* plane for neural interfacing. This matrix design not only enhances spatial resolution across the *xy* plane but also extends the coverage area, thereby increasing the effectiveness of neural interfacing. Importantly, these improvements are achieved without compromising the inherent qualities of microwires that make them suitable for long-term implantation. Additionally, this review will provide specific insights into the materials and techniques used to assemble individual electrodes in the *xy* plane, allowing the dura mater to be sutured over them. This approach is particularly relevant in light of recent reports by Neuralink,[18] which have shown that unsutured electrodes can be expelled by the brain—a problem that suturing the dura mater can help mitigate.

2 | NEURAL PROBES TAILORED TO THE BRAIN ARCHITECTURE

The nervous system consists of various distinct structures at different scales. In the central nervous system, axons can range in length from 100 microns to over 2 meters, with diameters varying from 0.2 to 20 microns.[19] Neuronal cell bodies, also known as somas, can have diameters ranging from 10 to 50 microns.[20] Specific neural circuits are challenging to define as they span various length scales, ranging from just a few neurons to macroscopic regions of the brain, spanning from sub-millimeter to centimeter scales. The complex topologies of neuronal and glial networks result from the dense interconnection of these diverse types of nerve cells. For example, in the mouse cerebral cortex, synapse density can be as high as one synapse per every 1.13 cubic microns.[21] In recent years, neural probes that mimic the interconnected structure of dendrite branches and approximate the diameters of neurites have been developed with the idea that at this scale, single neurons can be stimulated, and therefore higher resolution can be achieved in the way we communicate with the brain. Examples include the biomimetic probes, known as NeuE ribbons,[22] that imitate the natural neural scaffolds made up of radial glia, astrocytes, and the extracellular matrix (ECM). The inherent flexibility of these probes facilitates the migration and differentiation of neural progenitor cells while minimizing the activation of astrocytes and microglia. To date, commercially available microwire and silicon arrays utilized for the recording of single neuron activities have been found to result in substantial scar formation.[23–25] Extensive scarring results in a significant neuronal loss, leading to target neurons (for recording and/or stimulation) being farther from the electrode. Using electrodes that minimize

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brain damage enhances the signal-to-noise ratio, and it is essential for high-resolution recording.[24]

One critical aspect of biocompatibility is the prevention of neural inflammation, which can arise as a response to foreign objects in the brain, triggering a complex cascade of immune responses and compromising the functionality of the neural probes but, most importantly, the well-being of the patient.[26–28] Furthermore, assessing the longevity of these implants is essential to ensure their sustained effectiveness and the preservation of healthy neurons over time. Traditional rigid materials have been shown to trigger neural inflammation, leading to device failure.[28] To address this, researchers have developed ultrasoft electrodes with properties closer to brain tissue. Many studies have found that soft implants reduce inflammation in the brain and cause less cell body distortion compared to traditional metal implants, indicating their potential for long-term neural applications.[29–32]

Neural tissues exhibit mechanical properties that are softer than those of commercially available probes. Sub-cellular structures (i.e., axons and dendrites) are even more pliable.[33] Neural tissues experience regular motion caused by blood flow and periodic pressure changes due to heartbeat as well as movements of the brain within the skull during locomotion and head movement.[32,34] Mechanically, conventional neural probes such as silicon, metal-based, or flexible polyimide probes possess bending stiffnesses that are at least two to three orders of magnitude greater than that of brain tissue. For a brain slice with a thickness of 20–100 μm, the stiffness of brain tissue ranges from 10^{-4} to 10^{-1} nN m per unit width, while the stiffness of axons is several orders of magnitude smaller.[35] The disparities in size and mechanical properties between the probe and the brain are responsible for chronic inflammatory responses, including gliosis, scarring, and degradation of the neural interface. Moreover, the mechanical mismatch in stiffness leads to the displacement of the electrodes relative to the initial implantation site, making it challenging to monitor the same individual neurons and neural circuits over time.

3 | ELECTRODE RECORDING/ STIMULATING TIP PREPARATION

Due to the small size of the recording tips in microwires, the impedance of the core material is typically too high for low-noise recordings of spikes. Researchers have developed coatings that are able to increase the surface area of the recording sites, thus decreasing the impedance ($Z \propto 1/A$, where Z is the impedance magnitude and A is the surface area). Researchers' efforts in the material science field focus on the development of materials that can be locally deposited on the recording site and increase the surface area while providing long-term stability and small size of the implants.[36,37]

Typically, because of the conductive nature of the MwEAs, they also need to be coated with an insulating layer, such as parylene-C or alumina,[38,39] along the microwire side walls. This insulating layer is then removed at the tip of the electrodes to form the recording site. Due to the characteristic three-dimensionality of these arrays, removing the insulating layer in a reproducible manner from the tip is no trivial issue. A few techniques have been applied to remove parylene from the recording tips, including exposing them using fire, laser, and reactive ion etching.[37] A current challenge in MwEA fabrication is to consistently remove the same amount of insulating material so the recording sites are consistent between fabrication batches. Further, this insulating layer needs to be removed without damaging the core wire material. There may also be the need to apply an extra coating at the recording site to lower the impedance. An ideal coating material would reduce the impedance of the recording site below 100 kOhm, while not compromising its biocompatibility, long-term stability, and small size. These materials and their electrical and physical properties have been reviewed before.[36,37,40–42] A few popular materials include iridium oxide, platinum black, gold, and poly(3,4-ethylenedioxythiophene) (PEDOT). They are attractive because of their ability to be electrochemically deposited locally on the recording site and, hence, have good potential for large-scale manufacturing.

PEDOT is only suitable for long-term recording but not for stimulation applications, as the voltage applied for stimulating results in material degradation.[41] Furthermore, PEDOT:PSS has not demonstrated the ability to record and remain stable for periods longer than 6 months.[41] On the other hand, both iridium oxide and platinum black have been of increasing interest due to their ability to both record and stimulate. Coating the electrode tips with more conductive materials has been fundamental for the successful implementation of materials like carbon fibers. However, their long-term stability remains questionable with most studies in vivo lasting around 6 months. Electrode tips have not only been functionalized with conductive materials, but alternative materials such as proteins on the surface have also been tested to facilitate the insertion into the brain of very small wires. Further, protein coatings like lubricin have been used to modify the foreign body response and improve the long-term stability of the electrodes. By controlling the amount of proteins adsorbed on the material, the impedance of the electrodes remains stable for longer periods of time. Thus, it is expected that the use of proteins like lubricin will improve the longevity of electrode implants.[43] Liu et al.[31] added an anti-inflammatory drug to a polymer coating on a platinum/iridium microwire. The authors found that over a period of 7 days, the anti-inflammatory drug decreased inflammatory biomarkers. While fabrication, implantation, and core materials for MwEA are fundamental for the long-term use of implants, alternative coatings on the recording sites, or inclusion of proteins that mimic or cheat the brain into thinking the MwEA is not a foreign body, hold promise in the use of these implants, long-term.

4 | MICROWIRE ELECTRODE ARRAYS

MwEAs typically consist of two main components: (1) the microwire, designed for insertion into the brain and (2) the substrate that holds the wires into an array and facilitates the connection to external devices. Microwires are the core component of the MwEA,[32,44–46]

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TABLE 1 Materials commonly used for microwire array fabrication.

		Specifications				
	Design	Material	Dimensions	Impedance after coating (at 1 kHz)	Advantages	Disadvantages
Silicon-based	Utah Array[47]	Platinum or Iridium coated tips	Length: 0.5 to 1.5 mm Diameter: 10 μm Pitch: 400 μm	Platinum: 20-800 $k\Omega$ Sputtered iridium oxide: 1-80 $k\Omega$	Large-scale fabrication Reproducible fabrication Standard fabrication techniques	Stiff Glial scar formation due to stiffness mismatch
	Silicon microneedle array (SiMNA)[48]	Platinum nanomesh (PtNM) coated tips	Length: 100 μm Diameter: 10 μm Pitch: 50 to 400 μm	643 ± 208 $k\Omega$	Large-scale fabrication Reproducible fabrication Flexible	Short penetration of electrodes can lead to retraction of the implant
	Direct laser-written 3D electrodes[49]	Sputtered platinum-coated tips	Length: 350 μm Diameter: 20 μm Pitch: 90 μm	~200 kΩ	Reproducible fabrication Flexible	Short penetration of electrodes can lead to retraction of the implant
Metal-based	Tucker-Davis technologies (TDT) array[50,51]	Tungsten microwire	Length: 0.5 to 10 mm Diameter: 33 or 50 μm Pitch:375 μm	N/A	Conventional biocompatible materials Standard fabrication	Glial scar due to stiffness of microwire insertion challenges
	The ARGO system[38]	PtIr alloy (90% Pt/10% Ir) core/ uncoated tips	Length: 1 mm Diameter: 18 μm Pitch: 100–400 μm	300-500 $k\Omega$	Large-scale manufacturing Direct connection from microwire to external electronics Standard well-understood biocompatible metals High density	Potential insertion challenges due to dimpling with high density of wires
	3D Nanoprinted Array[52]	Silver core/PEDOT: PSS coated tips	Length: 1–2.5 mm Diameter: 30–100 μm Pitch: 200–300 μm	~237 $k\Omega$	Reproducible fabrication	Non-biocompatble silver core
	3D Expandable Microwire Electrode Arrays[53]	Nitinol wires/PEDOT: PSS-coated tips	Length: ~1 mm Diameter: 23 μm Pitch: 200–300 μm	~1.03 $M\ \Omega$	Flexible and conforming microwires	Potential challenge targeting specific areas Challenging insertion Specific manufacturing
Carbon-based	64-Channel Carbon Fiber Electrode Arrays[54]	Carbon fiber core/ PEDOT-TFB coated	Length: ~1 mm Diameter: Bundle of ~20 μm/ Carbon Fiber: 4.5 μm	170 ± 860 $k\Omega$	Biocompatible recording material	Thick bundles may lead to glial scar Nonbiocompatible silver paste for connection unsuitable for chronic recording Manual/discrete manufacturing
	High-density carbon fiber neural recording array technology[55]	Carbon fiber core/ PEDOT:PSS coated	Length: 1.4–3.5 mm Diameter: ~10 μm Pitch: 38 μm	~828 $k\Omega$	Biocompatible recording material	Nonbiocompatible silver paste for connection unsuitable for chronic recording Manual/discrete

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TABLE 1 (Continued)

Design	Specifications			Advantages	Disadvantages
	Material	Dimensions	Impedance after coating (at 1 kHz)		
Benchtop Carbon Fiber Microelectrode Array[56]	Carbon fiber core/ PEDOT:pTS coated	Length: 500 μm Diameter: 4.8–5.4 μm Pitch: 132–3000 μm	14.5 ± 1.3 kΩ	Biocompatible recording material Easy manufacturing	manufacturing Nonbiocompatible silver paste for connection unsuitable for chronic recording Manual/discrete manufacturing

and they can range from 8 μm to 50 μm thick. Usually, the recording sites sit either at the end or along the microwire core. A challenge for MwEAs is the ability of these microwires to be inserted into the brain with minimal damage. It is also desired that these wires have flexibility such that the array and electrodes can move in synch with brain micro-motions yet penetrate the brain with relative ease.[32] Usually, the substrate that holds the microwires is positioned on the surface of the brain cortex. The selection of the materials for both the microwire and substrate is paramount to ensure the long-term operation of MwEAs. The essential characteristics of these materials include biocompatibility, long-term stability, and adaptability for large-scale manufacturing. This section reviews the materials predominately used in microwires and substrates, exploring the properties vital for obtaining optimal, long-term MwEAs. Table 1 provides a summary of these common materials and their characteristics.

A prevailing challenge with MwEA is the complexity of their manufacturing. Given the small dimension of the microwires, generally the manufacturing process remains a manual endeavor, making the mass production of these arrays challenging. Establishing connections between individual minute wires and external devices for recording poses a significant challenge and potential point of failure during long-term implantation, particularly if the connection lacks robustness. Nonetheless, systems like the ARGO[38] and the configuration developed by Saleh et al.[52] have made progress. Both of these devices use standard manufacturing techniques to assemble high-density arrays. Overall, MwEAs can be classified into a layered or discrete wired design for assembly and external connection.[57] As illustrated in Figure 1a, the layered configuration features a printed circuit board (PCB) design, which provides robustness and serves as a platform to mount the connector and electrodes with predetermined spacing. However, the traces on the PCB limit the minimum spacing achievable between electrodes. Due to this constraint, the layered design is highly specific and dedicated to a limited number of applications. This approach requires high initial investment, with even and minor changes potentially incurring substantial costs, given the reliance on specialized manufacturing jigs. The advantage of this configuration is that the manufacturing processes are scalable and quality is highly controllable. Recent advances in flexible electronics allow for these types of device substrates to be flexible and made from more biocompatible materials.[48]

Figure 1b illustrates another array configuration, which joins the electrodes to wires and mechanically bonds the wires into a desired pattern.[57] This configuration can be achieved using a machined or 3D printed carrier or by potting with an adhesive into the desired shape. The free ends of the wires are routed to a connector or another PCB for external device connection. The discrete wire design can achieve tighter spacing and more complex patterns than a layered design in 1a. Furthermore, if designed correctly, this configuration can allow the dura to be sutured directly on top of the electrodes. However, the complexity of using non-standard materials as the substrate creates additional fabrication challenges. The discrete design is also generally less mechanically robust than the layered counterpart, which can lead to breaking of the microwires at the substrate joint.

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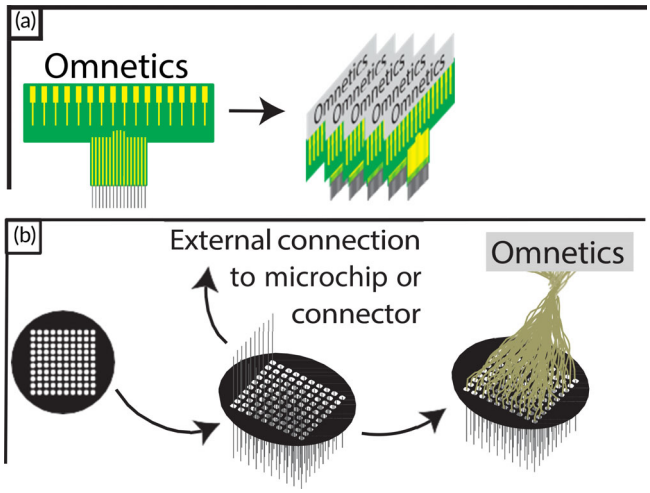

FIGURE 1 (a) Layered configuration showing discrete microwires attached to a printed circuit board (PCB) and subsequent layering of multiple PCBs to obtain an *xy* matrix. (b) Discrete configuration showing a substrate with discrete holes where microwires are to be inserted and later connected to an external PCB or connector.

Due to the intricate nature of MwEAs, they are generally manually assembled, which is highly labor-intensive. Although various alignment jigs and fixtures have been designed, the assembly process is highly dependent on an operator's skill, and the quality relies on the manual handling of the material.[58,59] This has represented a challenge in the commercialization of MwEAs. Here, we have classified MwEAs according to the material of the core microwire and identified three common classes: silicon-based, metal-based, and carbon-based microwires.

4.1 | Silicon-based

In the case of silicon-based arrays, no discrete microwires are used, enabling a more automated fabrication process. Fabrication of silicon arrays usually takes advantage of standard lithography techniques to etch away specific silicon shanks on the substrate.[60] The Utah array is the gold standard in MwEAs,[61] made out of silicon with standard microfabrication techniques. The fabrication processes are both scalable and reproducible.[16] These characteristics contribute to the successful commercialization of this array, and they have already been used both in clinical and animal research. A standard Utah consists of 100 electrodes tapered up to 1.5 mm long. The microwire needles start with 250 μm thick bases and finish with a diameter of around 5 μm at the tips.[62] These arrays are rigid, and the base of the needle is relatively thick. Ward et al. found that chronic damage to the tissue created by these arrays was significantly smaller compared to the MwEA developed by Tucker-Davis technologies—which consist of tungsten wires assembled in a discrete configuration—as shown in Figure 2a.[50] Nonetheless, recently, Patel et al. demonstrated that after 800 days of implantation, most neurons around the Utah array microwires were dead.[25] Moreover, the recording sites on the Utah array need to be coated with platinum or iridium oxide in order to reduce their impedances to a level suitable for electrophysiological recordings.[42] A recent study demonstrated that after 32 months of implantation, the electrode tips were damaged, and the amplitude of the recorded spikes decreased.[63] Many factors contribute to this failure, including delamination of the electrode tip and the broken microwire shanks. Furthermore, the Utah array's wires only vary from 0.5 mm to 1.5 mm, meaning when implanted in the brain, they reach superficial cortical layers.[63,64]

Lee et al. developed a silicon-based 1024-channel MwEA. This array was fabricated with standard lithography techniques, using a polyimide substrate for flexible backing. With only 10 μm thick silicon microwires, these electrodes did close to no damage to the brain, as demonstrated by their histology results (Figure 2b).[48] They demonstrated single-unit recording for up to 196 days. Attributed to the transparent nature of their flexible backing, they were able to image and provide optical stimulation through the back of the array. Nonetheless, these electrodes only have a length of 100 μm, which means that they can only record from a very superficial layer of the brain, so small movements of the array risk rejection from the brain. The flexible back end is beneficial, however, as conformity of penetrating arrays to the brain surface can improve the long-term performance of MwEA. There are concerns about their robustness and manufacturing reliably. This is because the interconnection of the microwires with the flexible back end is challenging and a potential failure mode.

Using two-photon lithography to 3D print non-conductive structures, Brown et al. developed a flexible MwEA.[49] The researchers created a 16-channel array with 250 μm-tall probes and 20 μm recording tips, spaced 90 μm apart. Similar to the device by Lee et al.,[48] the primary advantage of this design is its flexible backing. Additionally, the use of standard techniques enhances reproducibility and offers potential scalability. The researchers also reported that fabricating a 16-channel array takes approximately 10 minutes. Assuming linear scalability, this would extend to around 16 hours to fabricate a 1600-channel array, posing a challenge for commercialization.

In general, silicon-based MwEAs are attractive in commercialization due to low cost and the ease of scaling up manufacturing with standard techniques. Connecting these very small discrete microwires to the external world, however, remains challenging. New techniques that can improve the stability of the connection are demanding.

4.2 | Metal wire-based

Metals electrodes have been used for over 50 years.[41] Standard metal materials for electrodes include stainless steel, gold, tungsten, platinum, iridium, and derivatives due to their high electrical conductivity and good charge injection capacity, allowing both stimulation and recording.[40–42] Metal-based MwEAs using tungsten or platinum-iridium wires are typically assembled manually. In general, electrodes are manually placed into a holder with the desired pitch. The holder is often fabricated from silicon to take advantage of established microfabrication techniques.[53,57,65] Holders can either be vertically

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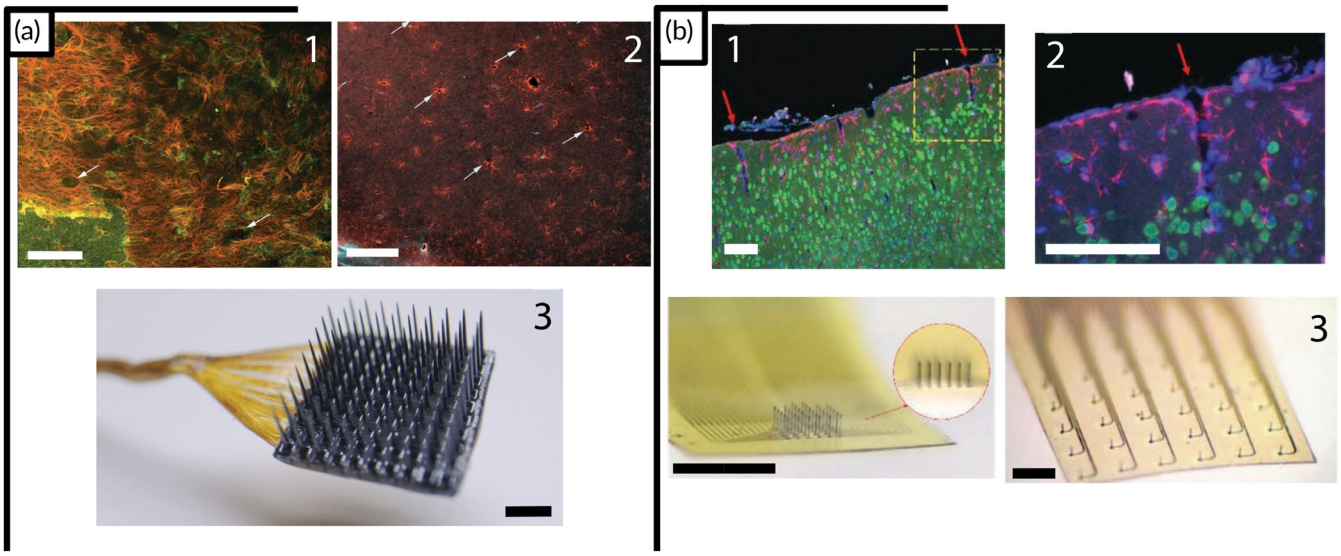

FIGURE 2 (a) (1 and 2) Histological image of rat brain implanted with a Utah array stained with anti-GFAP showing astrocytes (red), anti-beta-III tubulin showing neurons (green), and anti-CD68 showing macrophages/monocytes (light blue). Arrows point to holes caused by the shanks in the array. Scale bars represent 100 μm and 200 μm, respectively. With permission from.[50] (3) Optical image of a representative Utah array, the scale bar represents 900 μm. Taken from.[47] (b) (1 and 2) Histological image of a rat brain stained with DAPI showing cell nuclei (blue), NeuN antibody showing neurons (green), and Iba1 antibody showing microglia (magenta). Red arrows indicate penetration from a single shank into the tissue. Scale bars represent 100 μm. (3) Optical image of the flexible silicon-based array with 32 channels, scale bars represent 400 μm. With permission from.[48]

arranged where each electrode is inserted or horizontally arranged with each electrode laid down in individual grooves. They are secured in the holder using a biocompatible adhesive such as epoxy or silicone. Alignment jigs and tapes are designed to aid in the assembly of these devices. As the arrays are specific to each application, new assembly jigs are required and not transferable across designs. Furthermore, the use of metal microwires requires a compromise between thickness and stiffness. Thinner metal microwires that may produce less scarring when implanted become more bendable and thus are more difficult to implant, requiring the design of external jigs for implantation that may cause damage regardless.

Tucker-Davis technologies have commercialized a MwEA made of tungsten microwires,[51] which have a thickness of 33 μm and varying lengths from 0.5 mm to 2 mm. This MwEA has proven suitable for both in vivo stimulation and recording. Chronic performance has also been assessed.[50,66] Upon insertion, these rigid and relatively large electrodes tore the cortex, which led to high levels of scaring and tissue damage around the electrode long term, as shown in Figure 3a.[50] Ward et al. found that these electrodes only lasted for 30 days after implantation due to the high levels of glial scarring. Decreasing the diameters of the microwires may help reduce the glial scarring and enable longer-term recordings. Ideal microwires must have a stiffness similar to that of the brain to further reduce tissue damage due to mechanical mismatch. Likely due to the high scar formation of these electrodes; their use in research has reduced in recent years.

Sahasrabuddhe et al.,[38] recently developed the ARGO system, which utilizes platinum-iridium microwires as the electrode core, insulated with alumina. In this system, there are up to 1300 microwires, each 18 μm thick, 1 mm long, and with a varying pitch between 60 and 300 μm (Figure 3b). The single wires are directly connected to a complementary metal-oxide semiconductor (CMOS) chip on the back of the array by mechanically pressing the wires onto connections on the CMOS chip.[67] This high-density MwEA was able to penetrate the rat cortex and obtain excellent cortical recordings, and they were able to isolate 791 single units with a signal-to-noise ratio (SNR) of around 9. Microelectrode array development usually initiates with testing in small rodents, such as mice and rats. However, additional challenges are present when translating into humans as they possess thicker skulls, thicker dura layers, and larger relative tissue movement. Prior to translation, larger animal models like sheep and pigs are often utilized, offering a more accurate representation of the human brain surface. When testing the ARGO electrodes in sheep, the authors used surface electrodes instead of penetrating microwires to avoid damage upon insertion.[38] It is unclear whether the ARGO system can penetrate the brain surface, considering their large electrode count and the stiffness of the microwires. Furthermore, this very high density array has so far only been tested acutely. Further studies are required to understand the chronic tissue damage from high-density microwires. The primary advantage of this device is its high electrode density and the direct connection of individual electrodes to external electronics via a CMOS chip. To our knowledge, this device offers the highest electrode density in the *xy* plane. The direct connections enhance SNR by minimizing the distance signals travel before processing.

Saleh et al.[52] reported 3D printing of metal-based microwire arrays using an aerosol jet printer. In this work, the authors printed shanks with a tip diameter of 10 μm, length of 1 mm, and a spatial resolution of up to 2600 shanks/cm^2. This new fabrication technique can provide

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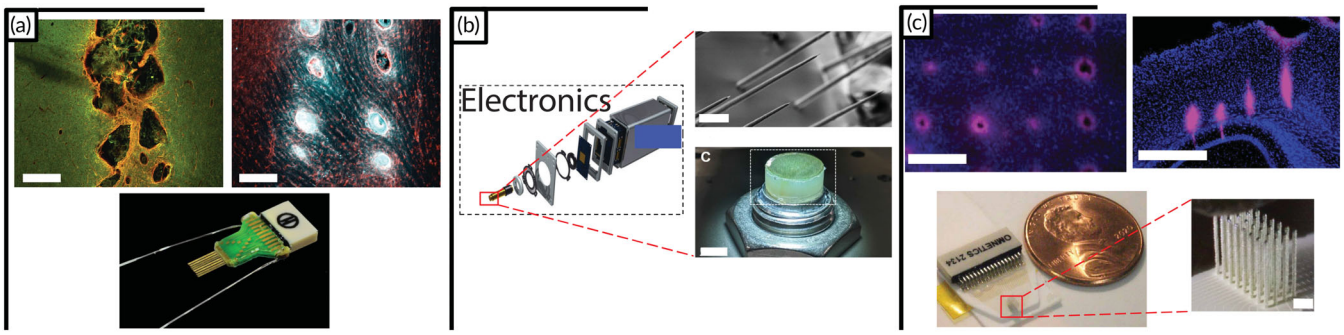

FIGURE 3 (a) Histological images of rat brain implanted with a Tucker–Davis technologies (TDT) array and stained with anti-GFPA showing astrocytes (red), anti-beta-III tubuli showing neurons (green), and anti-CD68 showing monocytes/macrophages (light blue). Scale bars represent 200 μm. With permission from,[50,51] (b) Block diagram and optical image of the ARGO system showing micrograph image of the single sharpened electrodes (scale bar: 100 μm) and optical macroscopic image of the array (scale bar: 5 mm). With permission from,[38] (c) (top) Histological image of rat brain after implantation with a 3D printed array after 30mins of implantation stained with DAPI showing cells nuclei (blue), and Evans blue showing cells with compromised membrane (red), with holes depicting the electrodes position. Scale bars represent 200 μm (bottom) Optical images of an 8 by 4 3D printed array; scale bar represents 500 μm.[52]

fast and custom prototyping of MwEAs, as well as, a solid interconnection between microwires and external devices. They demonstrated minimal damage upon insertion into the rat brain, as shown in Figure 3c. In the article, researchers used silver nanoparticles to fabricate the microwire shanks due to their low cost for prototyping. Silver is known to be toxic to humans[68]; thus, these arrays were not tested chronically. The authors, however, mentioned this technique can be translated to virtually any metal nanoparticle. This manufacturing technique has great potential for scalability if biocompatible particles are used. Recent developments are also aimed toward developing microelectrodes with shape memory; thus, they can be easily implantable and better conform to the brain. In this regard, Zhao et al.[53] developed a bundle of nitinol-based microwires that can be inserted into the brain and conform to the brain vasculature, generating minimal damage to the blood vessels upon implantation. In this design, the microwires are bundled into a needle and slowly inserted into the brain; as the electrodes are inserted, they return to their original shape and start conforming with the brain. The authors showed that this design does minimal damage upon insertion, which in turn is expected to improve the longevity of the implant.

4.3 | Carbon fiber-based

Carbon fibers have also been extensively used as microwires for MwEAs; their physical, mechanical, and chemical properties make them an ideal candidate for long-term recording. Their use in physiological recordings has been extensively studied.[30,37,54,55,69–72] Because of their small diameter, usually around 8 μm, carbon fibers lead to minimal damage to the brain upon insertion.[30,73] Figure 4a shows that after 12 weeks of implantation, there are plenty of remaining neurons at the electrode surface, suggesting the carbon fiber electrodes can interface with neurons, long-term, with minimal damage. Carbon fibers are expected to be able to record chronically with minimal degradation of the material, especially compared to other carbon-based materials such as carbon nanotube fibers or graphene fibers.[74] Their flexibility enables greater compliance to the brain tissue. This means that in addition to the little damage they cause upon implantation, they also have the potential of not causing damage due to any relative movement of the brain like other stiffer microelectrodes. However, to the best of our knowledge, chronic studies of the use of carbon fibers in vivo have not been longer than 4 months.[30,54,73,75,76]

Currently, high-density carbon fiber electrode arrays are assembled manually, which limits their scalability for high-volume production. Furthermore, to our knowledge, the only method used to connect the fibers to external electronics involves the use of silver paste, which is not biocompatible and highly limits the ability of carbon fibers to be used chronically in the clinic. To address the issue of manual handling, many researchers[58,77] have presented approaches in the automated handling and assembly of carbon fiber electrodes. In a system presented by Massey et al.[58], stock carbon fiber is passed through a roller feeder system with a glass capillary carrier. A camera outfitted with a position detection algorithm controls a motion system for alignment of the carbon fiber with the corresponding channel in the PCB. Once inserted, surgical scissors cut the carbon fiber from the feeder. Dong et al.[77] presented a similar system feeding carbon fiber electrodes into a silicon substrate, demonstrating the material's compatibility with automation in different substrates. A laser cutter system was implemented to improve reliability and consistency in cutting the carbon fibers to length.[77] Guitchounts et al.[54] presented a discrete wire array design to improve the handling and manufacturing throughput of carbon fiber-based MwEAs. As shown in Figure 4b, a 3D printed carrier consisting of 64 channels in a funnel arrangement, with the entry of the printed block matching the pitch of a commercially available connector and the exit of the block narrows down to an opening of 1 mm. The authors also report an improvement in operator handling time and ease of handling for subsequent tip processing steps. This approach still requires the manual insertion of individual fibers, and fibers are assembled and inserted in bundles, which increases tissue damage.[54] Richie et al.[56] show a practical method of fabrication for a 16-electrode carbon fiber device spaced with a

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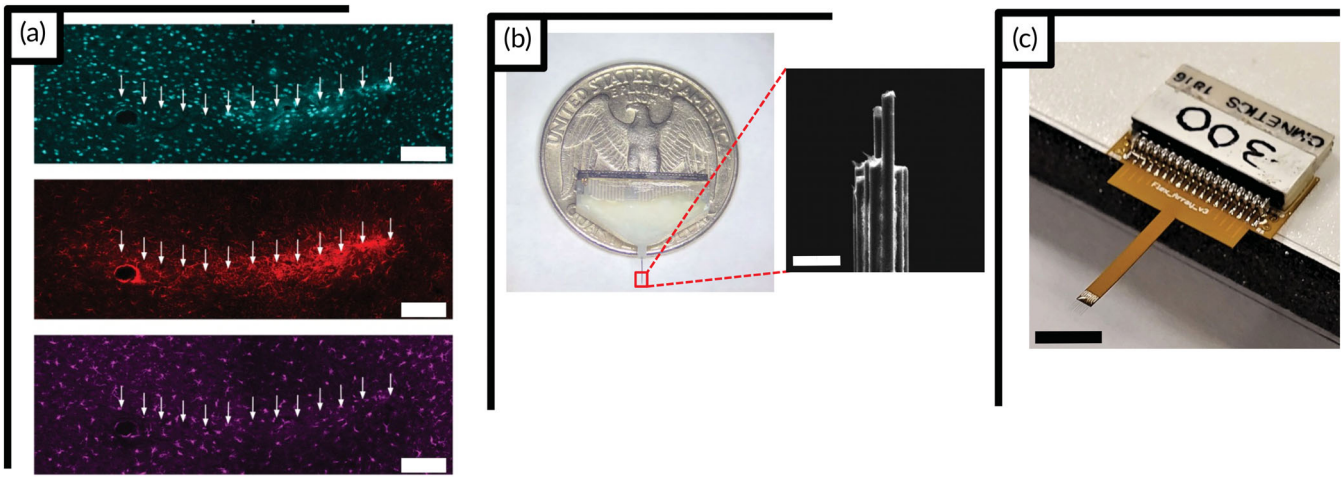

FIGURE 4 (a) Histological image of a rat brain implanted with a carbon fiber electrode array stained with NeuN depicting neurons (cyan), GFAP depicting astrocytes (red), and IBA1 depicting microglia (magenta); white arrows point to fiber tracks in the tissue. Scale bars represent 500 μm.[73] (b) Optical image and microscope image of the carbon fiber array with 3D printed array and hiroshi connector. Scale bar represents 20 μm.[54] (c) Optical image of flex array with carbon fibers at the tip and connected via an omnetics connector using custom-made printed circuit board. Scale bar represents 5 mm.[56]

100 μm pitch. The external connection includes an omnetics connector attached to a flex PCB (Figure 4c).[56] The device was tested for simultaneous single-unit recordings as well as neural stimulation.[78] This work showed the ability of the fibers to perform well in an acute setting, showing that carbon fiber electrodes are already practical as a commercial, acute research device.

Both Guitchounts' and Richie's designs utilize a layered configuration to connect with the carbon fibers, which limits their ability to sample in the *xy* plane. Additionally, they depended on a substrate in close contact with the fibers to facilitate insertion, further restricting the ability to suture the dura over the array. Massey et al. introduced the first carbon fiber array in a discrete configuration.[55] In their design, a silicon wafer was etched to have holes where the fibers could be threaded through. Traditional lithography techniques were used to connect the holes to external electronics, but the final connection to the carbon fiber used silver paste. Although this design represents significant progress toward high-density sampling using carbon fibers, the connection still employed silver paste, and the overall configuration of the final device makes suturing the dura over the array difficult. This array is a promising advancement in integrating carbon fibers with traditional lithography techniques while maintaining a discrete configuration.

For chronic assessment of carbon fibers, Patel et al.[30] compared the chronic unit activity from their 16 carbon fiber electrode array to that from a silicon shank electrode probes (Neuronexus). These showed that carbon fiber arrays outperformed the silicon probes over a period of up to 16 weeks.[30] The impedance values of the carbon fiber electrodes were consistent from 10 days after implantation up to 16 weeks. This implies that the coatings used (PEDOT:TFB) on the fibers showed promise for even longer-duration studies. These studies highlight the potential of carbon fiber-based MwEAs; however, the overall design of carbon fiber-based implants is still research-focused, and there is a long way until mass production can be done for commercialization. Carbon fiber-based MwEAs exhibit improved stability of coatings and treatments required for carbon fiber to be used as a neural electrode. However, improved substrate design for a reliable and practical chronic human implant is still required. Most of the current designs still use non-biocompatible products to make electrical connections to the carbon fiber, mainly silver paste. It is established that silver is toxic to the body, which limits the ability of these current arrays to be approved for human use.

5 | SURGICAL APPROACHES

Implantation remains another crucial factor influencing the longevity and chronic success of signal acquisition with MwEAs. Minimizing tissue damage during implantation and ensuring the implant's long-term integrity are critical considerations. Each of these can be mitigated by exploring and improving implant-specific electrode insertion, including mode and speed of insertion. We've previously highlighted the crucial role that dimensions and materials play in enhancing the longevity of implants. Here, we will discuss the impact of insertion and the consequent trauma experienced by the surrounding tissue during the implantation process.[79–82]

The invasive nature of penetrative electrodes requires the removal of layers of tissue and bone to expose the target tissue. These include the skin, periosteum, cranium, and, in some cases, the mater layers that sit above the brain (i.e., dura mater). These layers pose an increased challenge for more fragile or flexible MwEA as these are less likely to penetrate neural tissue.[83] For example, stiffer electrodes and greater insertion forces are required to penetrate into the dura mater.[84] To mitigate this, many implants require the removal of these layers during surgery. However, this adds complexity to the surgery as the dura mater can be difficult to remove without

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damaging the underlying tissue. A recent review by Thielen et al.[7] of insertion methods for general penetrating neural interfaces shows a variety of surgical techniques for specific implants. These range from the manual insertion of silicon probes, to pneumatic insertions of Utah arrays or the refined use of motorized drives for microwires. This represents another challenge in the widespread implementation of the MwEA, as surgeons need to be trained for very specifically designed insertion techniques.

The general force curve of a MwEA insertion is shown in Figure 5, and it involves the initial dimpling of the tissue, the penetration of the electrode through the surface, and then the frictional forces that increase over time as more of the electrode is embedded in the tissue.[85,86] The penetration force is particularly important as electrodes become smaller and more flexible materials are used. Neural implants made of stiff materials like tungsten and silicon, with elastic moduli typically over 100 GPa,[87] exhibit a steeper slope in the force-distance curve at the point of penetration and throughout the entire insertion process. With very rigid materials and a high Young's modulus, the large elastic modulus mismatch between the brain tissue (E = 0.4–15 kPa)[88] exacerbates friction-induced stress. In contrast, softer materials like polyimide and carbon fibers, with elastic moduli of 2.5–5 GPa,[89] show a shallower slope, meaning less force is required for displacement, making them potentially more suitable for neural implants as they are likely to cause less stress on the brain. Elastic materials can bend under force and return to their original shape; thus, after penetration, the force-distance curve rises gradually and plateaus quickly at larger displacements. If an electrode is unable to successfully penetrate the tissue, buckling will occur, wherein the electrode bends and any extra force merely adds to the bending meaning insertion is not possible. This buckling is a major limiting factor of microelectrode design and, hence, a determining factor on their viability as chronic implants. Buckling of electrodes can be mitigated through a number of different means, including the use of physical guides[53,90] and temporary stiffeners.[91,92] The issue with these interventions is that by adding mass and stiffness, there is increased acute damage incurred on the tissue compared to the bare microwires, which negates much of the benefit of using microelectrodes in the first place. These coatings and guides can also be difficult to apply reliably as well as being limited to specific configurations of electrodes. A review on the micro-motion of the brain and the insertion of such electrodes has previously been conducted by Sharafkhani et al.[32]

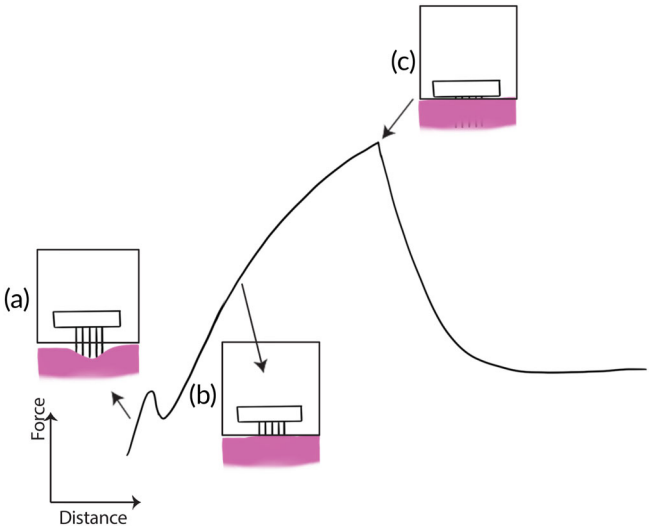

FIGURE 5 General force curve of a microelectrode array insertion showing force upon dimpling (a), insertion force (b), and end of insertion (c).

Another way to ease insertion of MwEA is to change their *xy* configuration and depth of the microwires. The shape of the implant will affect the initial penetration of the tissue. This concern diminishes as the implant size decreases but becomes more pronounced when dealing with high-density arrays. The shape affects the dimpling of the tissue in response to insertion and the ability of the implant to insert between neural cells with reduced strain on the surrounding tissue. Chen et al. explored the specific force profiles of dimpling, penetration, and frictional forces experienced by single tungsten electrodes with various tip geometries.[84] These experiments showed that the force required to rupture the pia and dura layers of the tissue was linearly related to electrode diameter, highlighting the benefit of microelectrodes over larger neural probes.[84] However, further work needs to be done to realize a solution that provides access to these microelectrodes into deeper neural structures in a manner that causes minimal damage to the tissue and the electrodes.

6 | FUTURE OUTLOOKS

Thanks to rapid advancements in machine learning, the decoding of single units from the brain's networks has become significantly faster and more efficient. As technology progresses, there arises a critical need to develop hardware and electrodes capable of interfacing seamlessly with the brain and computing technology over a lifetime. Microwires structured within a matrix present a promising solution for recording single units across various depths while ensuring adequate surface coverage, thereby facilitating the comprehensive mapping of networks in three dimensions. Moreover, the ability to customize dimensions and materials for these wires offers an opportunity to enhance the longevity of the implant. Thinner and more flexible wires, for instance, are anticipated to mitigate tissue scarring.[37]

A non-trivial challenge associated with microwire arrays lies in their fabrication and external connection. The assembly of single wires into the desired structure is typically performed manually, posing a barrier to scaling up manufacturing for commercialization. Recent advancements in flexible electronics, 3D printing, and lithography have enabled the rapid fabrication of microwire arrays.[48,52] However, these arrays often feature short microwires, typically in the range of tens of micrometers. This raises questions regarding the potential advantages over surface electrodes, as the penetration depth may not be substantial. Furthermore, the short length of the wires increases the likelihood of brain rejection, highlighting a significant concern.

Single microwires constructed from carbon or metal materials offer excellent conductivity, flexibility, and compact dimensions. However,

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the manual assembly of these individual wires into a matrix structure has presented significant challenges, often consuming considerable time and hindering scalability.[54,56] Recent efforts toward rapid manufacturing have emerged, such as Massey et al.'s development of a device capable of automatically threading carbon fibers through a substrate.[58] Additionally, Sahasrabuddhe et al. utilized sacrificial layers, primarily parylene-C, to arrange single metal microwires into an *xy* structure, enabling the swift assembly of up to 1000 microwires.[38]

The insertion method plays a crucial role in both the uptake and longevity of these arrays. Due to the high density and flexibility of microwire arrays, insertion can prove challenging and may elevate the risk of scar formation around the implant. Consequently, researchers have explored various techniques to mitigate dimpling and ensure reliable implantation.[7] Nonetheless, these solutions are often bespoke and vary between devices, posing an additional challenge for clinical adoption as physicians may require specialized training for each specific device.

The future of neural interfacing is directed toward the development of devices capable of long-term functionality post-implantation, while ensuring patient and tissue safety and delivering stable, high-resolution neural interfaces. As computational power continues to increase, there is a growing demand for more complex signals facilitated by a greater number of electrodes with minimal size. Recent advancements have shown effectiveness, particularly in short to medium-term neural recording within research settings. The overarching aim of neural engineers is to expand this technology to fully harness its potential and extend its application to therapeutics, allowing for lifelong implantation in human patients. This opens new avenues for the treatment and understanding of an unprecedented range of neurological disorders and pathologies.

AUTHOR CONTRIBUTIONS

S.E.D.: conceptualization, data curation, investigation, writing—original draft, and writing—review and editing. S.H.: conceptualization, data curation, investigation, and writing—review and editing. Y.J.J.: conceptualization, data curation, investigation, and writing—review and editing. W.T.: conceptualization, investigation, and writing—review and editing. D.J.G.: conceptualization, funding acquisition, and writing—review and editing.

ACKNOWLEDGMENTS

W.T. is supported by a Discovery Eary Career Researcher Award funded by the Australian Research Council (DE220100302). D.J.G. is supported by an ARC Future Fellowship (FT190100215).

CONFLICT OF INTEREST STATEMENT

S.H. is an employee of Carbon Cybernetics Group Ltd.; D.J.G. is a shareholder and office holder of Carbon Cybernetics Group Ltd., a company seeking to commercialize carbon fiber-based electrode array technologies; S.E.D., Y.J.J., and W.T. collaborate with Carbon Cybernetics.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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How to cite this article: De León SE, Higham S, Jung YJ, Tong W, Garrett DJ. Recent developments in microwire-structured intracortical electrode arrays for brain–machine interfaces. *Bioeng Transl Med*. 2024;e10742. doi:10.1002/btm2.10742

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